Lección 19

M. Siddikov

May 27, 2026

Plan de trabajo

- **Formulación hamiltoniana**
  - Transformaciones Canónicas
    - Evolución temporal = transformación canónica

- **Nuestro Syllabus:**

UNIDAD 5. Formulación Hamiltoniana
5.1. Transformaciones de Legendre
5.2. Ecuaciones de Hamilton
5.3. Ecuaciones de Hamilton de la Partícula Relativista
5.4. Corchete de Poisson
5.5. Transformaciones Canónicas
5.6. Ecuaciones de Hamilton-Jacobi
5.7. Teoría de perturbaciones

Summary of the last lecture

Hamilton's (Canonical) equations

$$\dot{p}_a = -\frac{\partial H(p, q, t)}{\partial q_a}, \qquad \dot{q}_a = \frac{\partial H(p, q, t)}{\partial p_a}, \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \qquad (1)$$

Canonical transformation: Transformation $(q, p, t) \to (Q, P, t)$ which preserves the form of canonical (Hamilton) equations, $i.e.$

$$\dot{P}_a = -\frac{\partial \tilde{H}(P, Q, t)}{\partial Q_a}, \qquad \dot{Q}_a = \frac{\partial \tilde{H}(P, Q, t)}{\partial P_a}, \qquad (1')$$

Requires sophisticated relation between derivatives:

$$\Rightarrow \qquad \left(\frac{\partial q_b}{\partial P_a}\right)_{Q,P} = -\left(\frac{\partial Q_a}{\partial p_b}\right)_{q,p}, \qquad \left(\frac{\partial Q_a}{\partial q_b}\right)_{p,q} = \left(\frac{\partial p_b}{\partial P_a}\right)_{Q,P}$$

The Poisson bracket is invariant, i.e. $[f, g]_{p,q} = [f, g]_{P,Q}$, so

$$[Q_a, Q_b]_{p,q} = 0, \quad [P_a, P_b]_{p,q} = 0, \quad [P_a, Q_b]_{p,q} = \delta_{ab},$$

Simplectic form is invariant,

$$\omega = \sum_a dq^a \wedge dp^a = \sum_a dQ^a \wedge dP^a$$

Summary of the last lecture (II)

$$\delta S = \delta \int dt \left(\sum_a p_a dq_a - H(p,\ q,\ t)\, dt \right) = \delta \int dt \left(\sum_a P_a dQ_a - \tilde{H}(P,\ Q,\ t)\, dt \right) = \min$$

$$\Rightarrow \left(\sum_a p_a dq_a - H(p,\ q,\ t)\, dt \right) = \left(\sum_a P_a dQ_a - \tilde{H}(P,\ Q,\ t)\, dt \right) + dF(q,\ Q,\ t)$$

►Can generate canonical transformations using generating functions $F(q,\ Q)$

$$\Rightarrow p_a = \frac{\partial F_1(q, Q)}{\partial q_a},\ P_a = -\frac{\partial F_1(q, Q)}{\partial Q_a}, \qquad \tilde{H} = H + \frac{\partial F_1}{\partial t} \qquad (1)$$

- Note that can rewrite

$$p_a dq_a = d(p_a q_a) - q_a dp_a$$

Let's introduce now

$$dF_2 = \sum q_a dp_a - \sum P_a dQ_a + \tilde{H}\, dt - H\, dt$$

$$F_2 = F_2(p,\ Q,\ t) = F_1 - \sum_a p_a q_a$$

⇒We could use F_2 as another generating function which depends on variables $p,\ Q,\ t$

$$P_a = -\frac{\partial F_2}{\partial Q_a}, \quad q_a = \frac{\partial F_2}{\partial p_a}, \quad \tilde{H} = H + \frac{\partial F_2}{\partial t}$$

Generating functions

► Can generate canonical transformations using generating functions $F_1 ... F_4$

$$\Rightarrow p_a = \frac{\partial F_1(q,Q)}{\partial q_a},\ P_a = -\frac{\partial F_1(q,Q)}{\partial Q_a}, \qquad \tilde{H} = H + \frac{\partial F_1}{\partial t} \tag{1}$$

$$\Rightarrow p_a = \frac{\partial F_2(q,P)}{\partial q_a},\ Q_a = \frac{\partial F_2(q,P)}{\partial P_a}, \qquad \tilde{H} = H + \frac{\partial F_2}{\partial t} \tag{2}$$

$$\Rightarrow q_a = -\frac{\partial F_3(p,Q)}{\partial p_a},\ P_a = -\frac{\partial F_3(p,Q)}{\partial Q_a}, \qquad \tilde{H} = H + \frac{\partial F_3}{\partial t} \tag{3}$$

$$\Rightarrow q_a = -\frac{\partial F_4(p,P)}{\partial p_a},\ Q_a = \frac{\partial F_4(p,P)}{\partial P_a}, \qquad \tilde{H} = H + \frac{\partial F_4}{\partial t} \tag{4}$$

For each $a = 1, ..., N$ should include: (1) either Q^a or P^a **and** (2) either q^a or p^a. All functions are linearly related by $F_i = F_j \pm P \cdot Q \pm p \cdot q$ $[d(PQ) = PdQ + QdP]$

– Sometimes may be very easy to work with one type of generating function but nearly impossible to use another (algebraic problems)

Example - 1

Find the generating function of the *identical* canonical transformation

$$q_a = Q_a, \quad p_a = P_a$$

Assume that corresponding generating function is $F_2\left(q,\, P,\, t\right)$

►Corresponding canonical transformations:

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a} = P_a, \;\; Q_a = \frac{\partial F_2}{\partial P_a} = q_a$$

Check that equations are compatible:

$$\frac{\partial^2 F_2}{\partial q_a \partial P_b} = \frac{\partial^2 F_2}{\partial P_b \partial q_a} = \delta_{ab}$$

⇒transformation is canonical (as expected), so can solve them together:

$$\Rightarrow F_2 = q_a P_a + \mathrm{const}$$

NOTE: If we try to find $F_1(q, Q)$ directly, can see that it is impossible:

−Have $q = Q$, but F_1 assumes that $q, Q$are independent variables (!?)

−Should express p and P as functions of q, Q, but it is impossible <u>algebraically</u>

Example 2

Find the values of α, β for which the transformation

$$q_a = \alpha Q_a, \quad p_a = \beta P_a$$

is canonical and find the generating function for this case

►Correponding canonical transformations:

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a} = \beta P_a, \;\; Q_a = \frac{\partial F_2}{\partial P_a} = \frac{1}{\alpha} q_a$$

Request that equations are compatible:

$$\frac{\partial^2 F_2}{\partial q_a \partial P_a} = \frac{\partial^2 F_2}{\partial P_a \partial q_a} \Rightarrow \beta = \frac{1}{\alpha}$$

$\Rightarrow$can solve them together for $\beta = 1/\alpha$:

$$\Rightarrow F_2 = \frac{1}{\alpha} q_a P_a + \mathrm{const}$$

Generating function

Find the generating function of the canonical transformation

$$Q = \log\left(\frac{\sin p}{q}\right), \quad P = q\cot p$$

Solution:
We already checked that transformation is Canonical, so focus on construction of generating function
−From definition of canonical transformation:

$$p \quad = \arcsin\left(q\,e^{Q}\right) = \frac{\partial F\,(q,\,Q,\,t)}{\partial q}, \quad P = \sqrt{e^{-2Q} - q^2} = -\frac{\partial F\,(q,\,Q,\,t)}{\partial Q}$$

Can check that these equations are compatible, evaluation of the second derivative $\partial^2 F/\partial q\partial Q$ from both gives the same result:

$$\frac{\partial^2 F\,(q,\,Q,\,t)}{\partial q\partial Q} = \frac{q\,e^{Q}}{\sqrt{1 - q^2 e^{2Q}}} \Rightarrow \text{system is compatible}$$

$$F \quad = \int dq\,\frac{\partial F\,(q,\,Q)}{\partial q} = q\arcsin\left(q\,e^{Q}\right) + \sqrt{e^{-2Q} - q^2}$$

Control problem

Demonstrate that the system of equations

$$\Rightarrow \frac{\partial F_1(q,Q)}{\partial q_a} = p_a(q,Q)\,, \quad \frac{\partial F_1(q,Q)}{\partial Q_a} = -P_a(q,Q)\,, \qquad (1)$$

always has solutions if $p_a(q,Q)\,,\ P_a(q,Q)$ are arbitrary functions which satisfy

$$\underbrace{\frac{\partial p_a(q,Q)}{\partial Q_b}}_{\partial^2 F/\partial q_a \partial Q_b} = -\underbrace{\frac{\partial P_b(q,Q)}{\partial q_a}}_{\partial^2 F/\partial Q_b \partial q_a}, \quad \underbrace{\frac{\partial p_a(q,Q)}{\partial q_b}}_{\partial^2 F/\partial q_a \partial q_b} = \underbrace{\frac{\partial p_b(q,Q)}{\partial q_a}}_{\partial^2 F/\partial q_b \partial q_a}, \quad \underbrace{\frac{\partial P_a(q,Q)}{\partial Q_b}}_{-\partial^2 F/\partial Q_a \partial Q_b} = \underbrace{\frac{\partial P_b(q,Q)}{\partial Q_a}}_{-\partial^2 F/\partial Q_b \partial Q_a}$$

i.e. if the transformation is canonical

Help: The best way to prove that the solution exists is to construct it explicitly

Control problem (solution)

Let's introduce variables ζ^a and functions ω^a

$$\zeta^a = \{q_1, ..., q_N, Q_1, ..., Q_N\}, \qquad \omega_a = \{p_1, ..., p_N, -P_1, ..., -P_N\}$$

$-$rewrite the system of equations as

$$\frac{dF_1}{d\zeta^a} = \omega_a(\zeta), \qquad [\text{diff. forms}: \boldsymbol{dF} = \boldsymbol{\omega}],$$

All compatibility conditions reduce to

$$\frac{\partial\omega_a}{\partial\zeta^b} - \frac{\partial\omega_b}{\partial\zeta^a} = 0, \qquad [\text{diff. forms}: \boldsymbol{d\omega} = 0]$$

$\Rightarrow$Need to recover the function from its gradient
$-$Solution (special case of Poincaré's theorem):

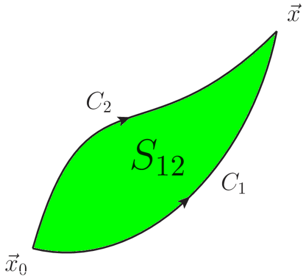
$$F_1(\zeta) = \int_0^\zeta d\xi^a \omega_a(\xi), \qquad F = \zeta^a \int_0^1 dt\, \omega_a(\zeta t)$$

where we choose to integrate over the straight line
*The change of lower limit only adds a constant to F_1, $F_1 \to F_1 + \text{const}$
*We have chosen straight line path from 0 to ζ^a, but the result integration does not depend on the choice of the path

Control problem (solution)

$$F_1(\zeta) = \int_0^{\zeta} d\xi^a \omega_a(\xi), \qquad F_1 = \zeta^a \int_0^1 dt\, \omega_a(\zeta t)$$

Lemma: The integral does NOT depend on the path:



Change of initial point⇒additive constant

$$\int_{x_0}^{x} = \int_{x_0}^{0} + \int_0^{x}$$

Proof:
Choose C_1, C_2 contours for comparison,

$$\int_{C_1} d\vec{\xi}\cdot\vec{\omega}\left(\vec{\xi}\right) - \int_{C_2} d\vec{\xi}\cdot\vec{\omega}\left(\vec{\xi}\right) = \oint_{C_{12}} d\vec{\xi}\cdot\vec{\omega}\left(\vec{\xi}\right)$$

► Stoke's theorem in N dimensions:

$$\int_{\partial S_{12}} \omega = \int_{S_{12}} \boldsymbol{d}\omega,$$

where $\partial S_{12} \equiv C_{12}$ is the contour (border) of S_{12}

► We have $\boldsymbol{d\omega} = 0$, so

$$\int_{\partial S} \omega = \int_S \boldsymbol{d}\omega = 0$$

Q.E.D.

Control question 2

Assume $\Phi(q, Q)$ and $F(q, Q) = -\Phi(Q, q)$ are two generating functions which correspond to canonical transformations T_1 and T_2 .

$-$**Example**: $F(q, Q) = q\,Q^2, \quad \Phi(q, Q) = -Q\,q^2$

Prove that these canonical transformations are inverse of each other, $T_1 = T_2^{-1}$

Control question 2 (proof)

Our transformation $p, q \to P, Q$ is defined by the generating functions $F(q, Q)$

$$\Rightarrow p_a = \frac{\partial F(q, Q)}{\partial q_a}, \;\; P_a = -\frac{\partial F(q, Q)}{\partial Q_a}, \tag{1}$$

Let's change letters $P, Q \leftrightarrow p, q$

$$\Rightarrow P_a = \frac{\partial F(Q, q)}{\partial Q_a}, \;\; p_a = -\frac{\partial F(Q, q)}{\partial q_a}, \tag{2}$$

Now recall that $F(q, Q) = -\Phi(Q, q)$

$$\Rightarrow P_a = -\frac{\partial \Phi(q, Q)}{\partial Q_a}, \;\; p_a = \frac{\partial \Phi(q, Q)}{\partial q_a}, \tag{3}$$

We got valid expression for generating function Φ which transform $p, q \to P, Q$. Since we permuted notations $P, Q \leftrightarrow p, q$, should identify Φ with inverse transformation $P, Q \to p, q$ in terms of original variables

Control question 3

In block 3 we discussed various infinitesimal symmetry transformations

$$\boldsymbol{q} \to \boldsymbol{q} + \delta\boldsymbol{q}, \qquad \boldsymbol{p} \to \boldsymbol{p} + \delta\boldsymbol{p}, \qquad \delta\boldsymbol{q} \ll q$$

which don't modify the action/hamiltonian. Demonstrate that the generating function

$$F_2 = \boldsymbol{q} \cdot \boldsymbol{P} + \epsilon\, G\left(\boldsymbol{q},\, \boldsymbol{P},\, t\right), \qquad \epsilon \ll 1$$

where $G(\boldsymbol{q},\, \boldsymbol{p},\, t)$ is the generator of transformation, corresponds to infinitesimal transformation

$$\delta q^a \equiv Q^a - q^a \approx \epsilon\,[G,\, q^a] + \mathcal{O}\left(\epsilon^2\right),$$
$$\delta p^a \equiv P^a - p^a \approx \epsilon\,[G,\, p^a] + \mathcal{O}\left(\epsilon^2\right),$$

Control question 3 (reply)

$$F_2 = \boldsymbol{q}\cdot\boldsymbol{P} + \epsilon\, G\left(\boldsymbol{q},\ \boldsymbol{P},\ t\right), \qquad \epsilon \ll 1$$

By definition we have

$$\Rightarrow p_a = \frac{\partial F_2(q,P)}{\partial q_a} = P_a + \epsilon \frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{P},\ t\right)}{\partial q^a},\ Q_a = \frac{\partial F_2(q,P)}{\partial P_a} = q_a + \epsilon \frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{P},\ t\right)}{\partial P^a} \tag{1}$$

For infinitesimal transformations ($\epsilon \ll 1$), can disregard the difference between P_a and p_a in $\mathcal{O}(\epsilon)$ terms:

$$\delta q^a \equiv Q^a - q^a = \epsilon \frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{P},\ t\right)}{\partial P^a} \approx \epsilon \frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{p},\ t\right)}{\partial p^a} + \mathcal{O}\left(\epsilon^2\right),$$

$$\delta p^a \equiv P^a - p^a = -\epsilon \frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{P},\ t\right)}{\partial q^a} \approx -\epsilon \frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{p},\ t\right)}{\partial q^a} + \mathcal{O}\left(\epsilon^2\right),$$

Can replace derivatives of G in $\mathcal{O}(\epsilon)$ terms using identities

$$\frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{p},\ t\right)}{\partial p^a} = \left[G,\ q^a\right], \qquad -\frac{\partial G\left(\boldsymbol{q},\ \boldsymbol{p},\ t\right)}{\partial q^a} = \left[G,\ p^a\right]$$

so end up with $\delta q^a \approx \epsilon\left[G,\ q^a\right], \delta p^a \approx \epsilon\left[G,\ p^a\right]$, Q.E.D.

Action angle variables

—Introduction

Harmonic oscillator: transformation to action-angle variables

Assume that the generating function is given by

$$F(q,\, Q) = \frac{m\omega q^2}{2} \cot Q$$

Find explicitly the canonic transformation which it generates and the hamiltonian of harmonic oscillator $H(P,\, Q)$ after this transformation

Caution: We'll see that transformation is NOT trivial and allows to drastically simplify analysis/solution of equations of motion

Harmonic oscillator: transformation to action-angle variables

$$\text{Hamiltonian}: \quad H = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}$$

$$\text{Transformation}: \quad p \quad = \frac{\partial F}{\partial q} = m\omega q \cot Q, \quad P = -\frac{\partial F}{\partial Q} = \frac{m\omega q^2}{2\sin^2 Q}$$

$$\text{"solve" algebraically to get } (q,p): \quad q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2m\omega P} \cos Q$$

Hamiltonian after transformation:

$$H_{\text{new}}(P,Q) = H_{\text{old}}\left(p(P,Q),\, q(P,Q)\right) + \underbrace{\frac{\partial F}{\partial t}}_{=0} = \omega\, P$$

-In new variables there is no dependence on Q at all, *i.e.* Q is cyclic variable !!!

$$\text{equations of motion}: \dot{P} = -\frac{\partial H_{\text{new}}}{\partial Q} = 0, \quad P = \text{const} = E/\omega$$

$$\dot{Q} = \frac{\partial H_{\text{new}}}{\partial P} = \omega \quad \Rightarrow Q = \omega t + \text{const}$$

Harmonic oscillator: transformation to action-angle variables

$$\text{Transformation}: \quad q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2m\omega P} \cos Q$$

$$\text{Trajectory}: \quad P = \text{const} = E/\omega, \quad Q = \omega t + \text{const} \qquad (1)$$

Compare with result (p, q) variables:

$$q = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \phi_0), \quad p = \sqrt{2mE} \cos(\omega t + \phi_0) \Rightarrow \text{agrees with } (1)$$

–Use of canonical transformation drastically simplified solution: instead of coupled ODEs

$$\dot{p} = -m\omega^2 q, \quad \dot{q} = p/m \qquad [\ddot{q} + \omega^2 q = 0]$$

solved MUCH simpler decoupled system (1a,1b).

*Such variables (P, Q) are called "action-angle" variables; please see Goldstein, Landau for more details.

**Exist for any system (even multidimensional)

**Drastically simplify analysis: always Q_a are cyclic:

$$\begin{aligned} H &= H(P_1, \ldots, P_N) \\ \dot{Q}_a &= \text{const} = \alpha_a \quad \Rightarrow \quad Q_a = \alpha_a t + \text{const} \end{aligned}$$

Action-angle variables (formal definitions)

Consider conservative system moving in potential field $U(q)$. Assume that the trajectory is closed, and define:

$$S_0\left(q,\, E\right) = \int^q p\, dq, \quad I\left(E\right) = \oint \frac{p\, dq}{2\pi}$$

$-S_0$ is the abbreviated action defined earlier
$-p$ is momentum. For conservative system $E = \mathrm{const}$, so

$$p = p(q, E) = \sqrt{2m(E - U(q))}$$

$-$Let's consider now $S_0\left(q,\, E\right) = S_0\left(q,\, I\right)$ as a generating function of a canonical transformation $F_2(q, P)$, in which I plays the role of a new canonical momentum P:

$$p = \frac{\partial F_2}{\partial q} = \frac{\partial S_0}{\partial q}, \quad Q \equiv w = \frac{\partial F_2}{\partial I} = \frac{\partial S_0}{\partial I}$$

$-$variable I is called *action*, w is *angle*

Action-angle variables (formal definitions)

Consider conservative system, assume that the trajectory is closed, and define:

$$S_0\,(q,\,E) = \int^q p\,dq, \quad I\,(E) = \oint \frac{p\,dq}{2\pi}, \quad Q \equiv w = \frac{\partial S_0}{\partial I}$$

- variable I is called <u>*action*</u>, w is <u>*angle*</u>
- Since Hamiltonian (energy E) depends only on canonical momentum but not on coordinate w, the variable w is cyclical, and

$$\dot{I} = -\frac{\partial H}{\partial w} = 0, \quad \dot{w} = \frac{dH}{dI} = \frac{dE}{dI} = \mathrm{const}$$

$$\Rightarrow w = \frac{dE}{dI}t + \mathrm{const}$$

We'll show now that the canonical transformation which we applied earlier to Harmonic oscillator is just transformation to action-angle variables.

Harmonic oscillator in action-angle variables

$$\text{abbreviated action(gener. function)}: \; S_0 = \int \sqrt{2m(\omega P - kq^2/2)}dq$$

$$p = \frac{\partial S_0}{\partial q} = \sqrt{2m\left(\omega P - \frac{k\,q^2}{2}\right)}, \qquad Q = \frac{\partial S_0}{\partial P} = \omega\frac{\partial S_0}{\partial E} = \arctan\left(\frac{\sqrt{k\,q^2}}{\sqrt{2\omega P - k\,q^2}}\right)$$

$$\Rightarrow \tan Q = \frac{\sqrt{k\,q^2}}{\sqrt{2\omega P - k\,q^2}} = \frac{q}{p}\sqrt{2km}, \quad q = \frac{p}{\sqrt{km}}\tan Q = \frac{p}{m\omega}\tan Q$$

$$\frac{p^2}{2m} = \left(\omega P - \frac{k\,q^2}{2}\right) = \left(\omega P - \frac{p^2}{2m}\tan^2 Q\right)$$

$$\frac{p^2}{2m}\left(1 + \tan^2 Q\right) = \frac{p^2}{2m\cos^2 Q} = \omega P$$

$$\Rightarrow q = \frac{p}{m\omega}\tan Q = \sqrt{\frac{2P}{m\omega}}\sin Q, \qquad p = \sqrt{2m\omega P}\cos Q,$$

in agreement with our previous findings

Evolution as canonical transformation

Evolution as canonical transformation

Assume that we wrote out canonical transformations at different time t_1 and t_2:

$$t_1: \quad \dot{q}_1 = \frac{\partial H\left(p_1,\, q_1\right)}{\partial p_1}, \quad \dot{p}_1 = -\frac{\partial H\left(p_1,\, q_1\right)}{\partial q_1}, \qquad p_1 \equiv p\left(t_1\right), \quad q_1 \equiv q\left(t_1\right)$$

$$t_2: \quad \dot{q}_2 \quad = \frac{\partial H\left(p_2,\, q_2\right)}{\partial p_2}, \quad \dot{p}_2 = -\frac{\partial H\left(p_2,\, q_2\right)}{\partial q_2}, \qquad p_2 \equiv p\left(t_2\right), \quad q_2 \equiv q\left(t_2\right)$$

In the process of evolution, the system changes its momenta and coordinates continuously.

$-$Since canonical equations are valid both at t_1 and t_2, conclude that evolution $(p_1, q_1) \to (p_2, q_2)$ is some canonical transformation

Action as a function of endpoint coordinates

Up to now we introduced action assuming that the trajectory is fixed at endpoints. Let's analyze now how S depends on these coordinates:

$$S \equiv \int_{q_i, t_1}^{q_f, t_2} dt\, L\left(q_a,\, \dot{q}_a\right) = \int_{q_i, t_1}^{q_f, t_2} dt\, \left(p_a \dot{q}_a - H\left(q_a, p_a\right)\right) \quad (1)$$

– Action S is a **functional** of possible trajectories, but is a function of endpoint values $q_a^{(i)}(t_1),\ q_a^{(f)}(t_2)$
(we use superscript (f) for final coordinates in the moment t_2 and (i) for initial coordinates in the moment t_1)

– Assume that $q_{\mathrm{cl}}(t)$ is a solution of Euler-Lagrange equation with given initial and final conditions. If we substitute it into (1), it turns into just a function of coordinates:

$$\Rightarrow S = S\left(q_a^{(f)} t_2,\ q_a^{(i)},\, t_1\right)$$

Example

Evaluate explicitly the action $S\left(q_a^{(f)} t_2,\ q_a^{(i)}, t_2\right)$ for the case of a free particle and harmonic oscillator.

–For a **free particle** $q(t) = q_0 + vt$, so using initial conditions to fix q_0, v, can immediately get

$$q(t) = q_i + \frac{q_f - q_i}{t_2 - t_1}(t - t_1)$$

$$S_{\text{free}} = \int_{q_i, t_1}^{q_f, t_2} dt\, \frac{m\dot{q}^2}{2} = \frac{m\,(q_f - q_i)^2}{2\,(t_2 - t_1)}$$

–For a particle in the **harmonic oscillator** potential

$$q(t) = A \sin\omega t + B\cos\omega t = \mathcal{A}\sin\omega\,(t - t_1) + \mathcal{B}\cos\omega\,(t - t_1)$$

using initial conditions can fix $\mathcal{A}$, $\mathcal{B}$ and get

$$q(t) = \frac{q_f - q_i\cos\omega T}{\sin\omega T}\sin\omega\,(t - t_1) + q_i\cos\omega\,(t - t_1)\,, \qquad T = t_f - t_i.$$

$$S_{\text{oscillator}}\left(q_f, t_2;\ q_f, t_1\right) = \int_{q_i, t_1}^{q_f, t_2} dt\,\left[\frac{m\dot{q}^2}{2} - \frac{m\omega^2 q^2}{2}\right] =$$

$$= \frac{m\omega}{2\sin(\omega T)}\left(\left(q_i^2 + q_f^2\right)\cos(\omega T) - 2q_f q_i\right).$$

Action as a function of endpoint coordinates

Up to now we introduced action assuming that the trajectory is fixed at endpoints. Let's analyze now how S depends on these coordinates:

$$S \quad \equiv \int_{q_i,t_1}^{q_f,t_2} dt\, L\left(q_a,\, \dot{q}_a\right) = \int_{q_i,t_1}^{q_f,t_2} \left(p_a dq_a - H\left(q_a, p_a\right) dt\right) \qquad (1)$$

−Assume that $q_{\mathrm{cl}}(t)$ is a solution of Euler-Lagrange equation with given initial and final conditions. If we substitute it into (1), it turns into just a function of coordinates:

$$\Rightarrow S = S\left(q_a^{(f)} t_2,\, q_a^{(i)}, t_2\right)$$

−Let's take differentials of initial/final coordinates:

$$\Rightarrow dS = \sum p_a^{(f)} dq_a^{(f)} - \sum p_a^{(i)} dq_a^{(i)} + \left(H^{(i)} - H^{(f)}\right) dt$$

Derivatives of action:

$$\frac{\partial S}{\partial q_a^{(f)}} = p_a^{(f)}, \quad \frac{\partial S}{\partial q_a^{(i)}} = -p_a^{(i)}, \qquad \frac{\partial S}{\partial t^{(f)}} = -H^{(f)}, \quad \frac{\partial S}{\partial t^{(i)}} = H^{(i)}$$

Evolution as canonical transformation

$$dS = \sum p_a^{(f)} dq_a^{(f)} - \sum p_a^{(i)} dq_a^{(i)} + \left(H^{(i)} - H^{(f)}\right) dt$$

Compare this with expression for generating function introduced earlier:

$$dF = \sum p_a dq_a - \sum P_a dQ_a + \left(\tilde{H} - H\right) dt$$

$\Rightarrow$Can interpret $S\left(q, \; Q, \; t\right)$ as generating function of this transformation!

Liouville's theorem

Liouville's theorem

1) The canonical transformations do not change the phase volume element

$$dp_1 ... dp_N dq_1 ... dq_N = dP_1 ... dP_N dQ_1 ... dQ_N,$$

i.e. the Jacobian of canonical transformation is always $= 1$.

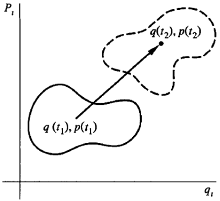
2) The evolution of the system does not change the phase volume element

$$\begin{aligned} d\Omega &\equiv dp_1(t_1) ... dp_N(t_1)\, dq_1(t_1) ... dq_N(t_1) \\ &= dp_1(t_2) ... dp_N(t_2)\, dq_1(t_2) ... dq_N(t_2) \end{aligned}$$

Since evolution is a canonical transformation with generating function given by its action S, then it is clear that statement (2) directly follows from the statement (1).

Liouville's theorem - illustration

Illustration:



The integral $\int d\Omega$ over a finite volume (solid→dashed contour) does not change during evolution

Assume at $t = 0$ You measured momenta and coordinates of particle with errors Δp, Δq. In phase space, You may consider that "true" momenta and coordinates are distributed in some small region $\Delta p \times \Delta q$. In the process of evolution, at later moment t errors $\Delta p_t, \Delta q_t$ will be different. However, can see that Liouville's theorem imposes constraints on possible $\Delta p_t, \Delta q_t$

Instead of single particle, You may consider ensemble of N particles and analyze phase space distribution $d\rho/d\Omega$ as a function of time.

Liouville's theorem (proof)

Prove that canonical transformations do not change the phase volume element

$$d\Omega \equiv dp_1 ... dp_N dq_1 ... dq_N$$

We need to show that the Jacobian of transformation from old to new coordinates

$$J = \frac{D\left(p_1 ... p_N,\ q_1 ... q_N\right)}{D\left(P_1 ... P_N,\ Q_1 ... Q_N\right)} = 1$$

under canonical transformation (clearly, this is necessary and sufficient condition).
−**Case of 1 dimension**:

$$J = \left| \frac{\partial p}{\partial P} \frac{\partial q}{\partial Q} - \frac{\partial p}{\partial Q} \frac{\partial q}{\partial P} \right| = [p,\ q]_{P,Q} = 1$$

Liouville's theorem (proof)

Case of N dimensions:

—We will consider our transformation as a superposition of 2 steps: $(p, q) \to (P,\ q) \to (P, Q)$

*While transformation $(p, q) \to (P, Q)$ is canonical, individual steps may be not canonical at all. However, for evaluation of jacobian this is not important

*For a product (transposition) of two transformations, the jacobian J equals the product of jacobians, $J = J_1 J_2$,

$$J = \frac{D\left(p_1 ... p_N,\ q_1 ... q_N\right)}{D\left(P_1 ... P_N,\ Q_1 ... Q_N\right)} = \frac{D\left(p_1 ... p_N,\ q_1 ... q_N\right)}{D\left(P_1 ... P_N,\ q_1 ... q_N\right)} \frac{D\left(P_1 ... P_N,\ q_1 ... q_N\right)}{D\left(P_1 ... P_N,\ Q_1 ... Q_N\right)}$$

Evaluation of Jacobians:

$$\frac{D\left(p_1 \ldots p_N,\ q_1 \ldots q_N\right)}{D\left(P_1 \ldots P_N,\ q_1 \ldots q_N\right)} \equiv \det \begin{vmatrix} \frac{\partial p_1}{\partial P_1} & \ldots & \frac{\partial p_N}{\partial P_1} & 0 & 0 & 0 \\ \vdots & \ddots & \vdots & 0 & 0 & 0 \\ \frac{\partial p_N}{\partial P} & \ldots & \frac{\partial p_N}{\partial P} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \ldots & 0 \\ 0 & 0 & 0 & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \ldots & 1 \end{vmatrix} =$$

$$= \det \begin{vmatrix} \frac{\partial p_1}{\partial P_1} & \ldots & \frac{\partial p_N}{\partial P_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial p_1}{\partial P_N} & \ldots & \frac{\partial p_N}{\partial P_N} \end{vmatrix} = \left.\frac{D\left(p_1 \ldots p_N\right)}{D\left(P_1 \ldots P_N\right)}\right|_{Q=\mathrm{const}}$$

– When we evaluate Jacobian, we effectively may disregard (omit) variables $q_1 \ldots q_N$ which are the same before and after transformation

Liouville's theorem (proof)

$$J = \frac{D\left(p_1 \ldots p_N,\ q_1 \ldots q_N\right)}{D\left(P_1 \ldots P_N,\ Q_1 \ldots Q_N\right)} = \frac{D\left(p_1 \ldots p_N,\ q_1 \ldots q_N\right)}{D\left(P_1 \ldots P_N,\ q_1 \ldots q_N\right)} \frac{D\left(P_1 \ldots P_N,\ q_1 \ldots q_N\right)}{D\left(P_1 \ldots P_N,\ Q_1 \ldots Q_N\right)} =$$

$$= \left.\frac{D\left(p_1 \ldots p_N\right)}{D\left(P_1 \ldots P_N\right)}\right|_{Q=\mathrm{const}} \cdot \left.\frac{D\left(q_1 \ldots q_N\right)}{D\left(Q_1 \ldots Q_N\right)}\right|_{P=\mathrm{const}}$$

But we have seen that for the canonical transformations

$$\left(\frac{\partial Q_a}{\partial q_b}\right)_{p,q} = \left(\frac{\partial p_b}{\partial P_a}\right)_{Q,P}$$

Let's consider this as matrix equation and take $\det(\ldots)$ of both parts:

$$\Rightarrow \left.\frac{D\left(p_1 \ldots p_N\right)}{D\left(P_1 \ldots P_N\right)}\right|_{Q=\mathrm{const}} = \left.\frac{D\left(Q_1 \ldots Q_N\right)}{D\left(q_1 \ldots q_N\right)}\right|_{Q=\mathrm{const}} = \left(\left.\frac{D\left(q_1 \ldots q_N\right)}{D\left(Q_1 \ldots Q_N\right)}\right|_{P=\mathrm{const}}\right)^{-1}$$

$\Rightarrow$ conclude that

$$J = \left.\frac{D\left(p_1 \ldots p_N\right)}{D\left(P_1 \ldots P_N\right)}\right|_{Q=\mathrm{const}} \cdot \left.\frac{D\left(q_1 \ldots q_N\right)}{D\left(Q_1 \ldots Q_N\right)}\right|_{P=\mathrm{const}} = 1, \qquad \text{Q.E.D.}$$

Liouville's theorem in symplectic notations

Proof:
1) Define the volume form in $2n$-dimensional phase space

$$\sigma = \underbrace{\omega \wedge \ldots \wedge \omega}_{n} \sim dq_1 \wedge \ldots \wedge dq_n \wedge dp_1 \wedge \ldots \wedge dp_n$$

$-$Since $\omega = dq^a \wedge dp^a$ (2-form), $\omega \wedge \omega \neq 0$:
*In general if A and B are p-form and q-form, then

$$A \wedge B = (-1)^{pq}\, B \wedge A$$

**If A and B are 1-forms, then $A \wedge B = -B \wedge A$ and $A \wedge A = B \wedge B = 0$
**If at least one of the A or B is of even order (2,4,...) then $A \wedge B = B \wedge A$ and $A \wedge A \neq 0$

Liouville's theorem in symplectic notations

Proof:

1) Define the volume form in $2n$-dimensional phase space

$$\sigma = \underbrace{\omega \wedge \ldots \wedge \omega}_{n} \sim dq_1 \wedge \ldots \wedge dq_n \wedge dp_1 \wedge \ldots \wedge dp_n$$

$-$Since $\omega = dq^a \wedge dp^a$ (2-form), $\omega \wedge \omega \neq 0$

$-$We've shown in the last lecture that

$^*\omega$ is invariant under canonical transformation, so σ is also invariant

**Mathematics: All canonical transformations which keep skew-symmetric form form a group $\mathrm{Sp}\,(2n, \mathbb{R})$

$$\forall M \quad \in \mathrm{Sp}\,(2n, \mathbb{R})\,, \quad M^T \Omega M = \Omega,\ \Omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$$

$$\forall \xi, \eta, \quad [M\boldsymbol{\xi},\ M\boldsymbol{\eta}] = [\boldsymbol{\xi},\ \boldsymbol{\eta}]$$

**Jacobian of any symplectic transform $J = 1$

*Under evolution

$$\mathcal{L}_U \omega = 0,$$

so applying Leibnitz rule, conclude that $\mathcal{L}_U \sigma = 0.$, which proves that phase volume remains invariant

Corollary

$-$Under evolution

$$\mathcal{L}_U\omega = 0,$$

$$\Downarrow$$

Any number of products $\omega \wedge \omega, \quad \omega \wedge \omega \wedge \omega, \ldots$

$$\sigma_k = \underbrace{\omega \wedge \ldots \wedge \omega}_{k<n}$$

remain invariant under evolution and/or canonical transformation

Summary

- For any canonical transformation there is at least one a generating function of one of the types $F_1 ... F_4$ which generates it.
- The canonical transformations can drastically simplify Hamiltonian
  - Action-angle variables: all coordinates become cyclic
- The time evolution is also a canonical transformation
  - The generating function is action calculated along the classical trajectory.

$$\underbrace{dS}_{dF_1} = \sum \underbrace{p_a^{(f)} dq_a^{(f)}}_{p_a dQ_a} - \sum \underbrace{p_a^{(i)} dq_a^{(i)}}_{P_a dQ_a} + \underbrace{\left(H^{(i)} - H^{(f)}\right)}_{\tilde{H} - H} dt$$

- The canonical transformations (and time evolution) do not change the phase volume

$$dq_1 \wedge ... \wedge dq_n \wedge dp_1 \wedge ... \wedge dp_n$$